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Article *in* Journal of Global Antimicrobial Resistance · June 2023

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Journal of Global Antimicrobial Resistance 34 (2023) 18–22

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Short Communication

Molecular characterization and antimicrobial resistance of nasal *Staphylococcus aureus* in the community of Kabul

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ARTICLE INFO

Article history:
Received 6 October 2022
Revised 24 March 2023
Accepted 4 June 2023
Available online 14 June 2023

Editor: Prof Geoffrey Coombs

Keywords:
Nasal carriage
MRSA clones
Antibiotic resistance
Molecular typing
Kabul

ABSTRACT

Objectives: This study aimed to investigate the prevalence and molecular characteristics of community methicillin-resistant *Staphylococcus aureus* (MRSA) nasal carriage among students at Kabul University.
Methods: Nasal swabs were collected from anterior nares of 150 healthy non-medical students at Kabul University. Antimicrobial susceptibility testing was performed on all *S. aureus* isolates, and all detected MRSA isolates were then confirmed by *mec*A/*mec*C polymerase chain reaction and characterized using DNA microarray.
Results: A total of 50 *S. aureus* strains were isolated from the anterior nares of the 150 participants. The prevalence of *S. aureus* and MRSA nasal carriage among Kabul students was 33.3% and 12.7%, respectively. Seven (36.8%) MRSA isolates and 8 (25.8%) methicillin-susceptible *S. aureus* (MSSA) isolates were multidrug-resistant (i.e. resistant to at least three different antimicrobials tested). All MRSA isolates (n = 19) were susceptible to linezolid, rifampicin, and fusidic acid. Seven MRSA clones, belonging to four clonal complexes (CCs), were identified. The most commonly identified clone was CC22-MRSA-IV TSST-1-positive, which accounted for 63.2% (12/19) of MRSA isolates. SCC*mec* typing showed that most MRSA strains harboured SCC*mec* type IV (94.7%). Thirteen (68.4%) MRSA isolates carried the TSST-1 and 5 (26.3%) PVL genes.
Conclusion: Our findings revealed the relatively high prevalence of MRSA nasal carriers in the community in Kabul, with the predominance of the CC22-MRSA-IV TSST-1-positive clone and frequent multidrug resistance among these isolates.
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1. Introduction

Since methicillin-resistant *Staphylococcus aureus* (MRSA) was first identified in 1961, it has been associated with higher mortality rates and increased lengths of hospital stays as well as associated health care costs [1]. MRSA infections are a serious risk for patients in healthcare settings and a challenge for public health, whereas infections due to community-acquired MRSA (CA-MRSA) seem to be increasing among people in different areas of the world [2]. Nasal carriage of *S. aureus* has been shown to play a key role in the pathogenesis of *S. aureus* infections, particularly in patients undergoing surgery, dialysis, and/or in intensive care units [3]. It also constitutes a risk factor for the development of skin and soft tissue infections in non-hospitalized and non-diseased individuals [4]. MRSA strains that reside in the anterior nares constitute a reservoir that predisposes the host to future infections [5]: in a systematic review, patients colonized with MRSA were four times more likely to develop invasive infection than patients colonized with MSSA [6]. Therefore, determining the prevalence of *S. aureus* nasal carriage—as well as antimicrobial resistance profiles and molecular typing of nasal MRSA isolates in healthy populations—is useful for understanding the epidemiology of community-acquired infections and supporting infection control measures.

In central Asia, including Afghanistan, studies of MRSA have mainly focused on the prevalence of MRSA, but few data are avail-

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https://doi.org/10.1016/j.jgar.2023.06.001
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Table 1
Antimicrobial resistance rates of *S. aureus*, MRSA, and MSSA nasal isolates among non-medical students in Kabul, Afghanistan, between March and July 2018 (N = 50).

Antibiotic	Resistance rate (%)			
	S. aureus (N = 50)	MRSA (n = 19)	MSSA (n = 31)	*P* value
Penicillin G	96	100	93.5	0.26
Erythromycin	32	36.8	29	0.57
Clindamycin	8	10.5	6.5	0.61
Norfloxacin	18	26.3	12.9	0.23
Kanamycin	30	26.3	32.3	0.66
Tobramycin	22	10.5	29	0.13
Gentamicin	6	10.5	3.2	0.29
Cotrimoxazole	18	15.8	19.4	0.75
Linezolid	0	0	0	-
Fusidic acid	4	0	6.5	0.26
Chloramphenicol	2	5.3	0	0.2
Fosfomycin	8	5.3	9.7	0.58
Rifampicin	0	0	0	-

able about circulating clones. A recent study concerning *S. aureus* infections diagnosed in Kabul hospitals highlighted a high rate of methicillin resistance (66.3%) and the presence of multiple virulence and antibiotic resistance genes in *S. aureus* isolates, with a predominance of CC30-MRSA-IV PVL-positive, CC22-MRSA-IV TSST-1-positive, and ST772-MRSA-V PVL-positive clones [7]. However, no data are available about the prevalence of MRSA carriage and the clones circulating in the community.

The present study aimed to investigate the prevalence of MRSA nasal carriage and the molecular characteristics of MRSA isolates in the community of Kabul through *S. aureus* nasal screening in non-medical students at Kabul University.

2. Materials and methods

2.1. Study population

A total of 150 healthy student volunteers from non-medical faculties of Kabul University were enrolled in this study from March to July 2018. Informed consent was obtained from all participants included, according to Afghanistan ethical rules. Every student with a history of antibiotic consumption, hospitalization, admission to a nursing home, or surgery during the last 2 months were excluded.

2.2. Identification of S. aureus *isolates and antimicrobial susceptibility testing*

Using sterile moistened swabs, samples were collected from both anterior nares of volunteers. Standard microbiological procedures were conducted on nasal swabs with minimum delay for culture. Samples were cultured for 18–24 hours at 37°C onto blood agar base medium (Oxoid, Basingstoke, United Kingdom) supplemented with 5% sheep blood. Then presumptive *S. aureus* isolates were identified by carrying out confirmatory tests, consisting of a Pastorex Staph Plus latex agglutination test (Bio-Rad, Marnes-la-Coquette, France) and tube coagulase assay (EDTA-rabbit plasma; Difco Laboratories, Detroit, MI, US).

Antimicrobial susceptibility testing was performed on Mueller Hinton agar (Oxoid) by Kirby Bauer disc diffusion method (antibiotic discs from i2a, Montpellier, France) according to the 2019 guidelines of the European Committee on Antimicrobial Susceptibility Testing [8]. The following antimicrobial agents were tested: penicillin G, cefoxitin (used to detect methicillin resistance), erythromycin, clindamycin, norfloxacin, kanamycin, tobramycin, gentamicin, cotrimoxazole, linezolid, fusidic acid, chloramphenicol, fosfomycin, and rifampicin.

2.3. Detection of mec*A*/mec*C genes and* agr *typing by multiplex PCR*

Cellular DNA was obtained from *S. aureus* colonies grown overnight on blood agar plates using a DNA Extraction Kit (Promega, Madison, WI) in accordance with manufacturer's instructions. The detection of the *mec*A and *mec*C genes and *agr* typing were performed by multiplex polymerase chain reaction using primers already published [9–11].

2.4. Molecular characterization of strains by DNA microarray

The DNA microarray Identibac *S. aureus* genotyping® (Alere Technologies, Jena, Germany) was used for MRSA isolates as previously described [12]. This microarray allows the detection of 336 different target sequences corresponding to 185 genes and their allelic variants, including both virulence and resistance genes. The assignation of isolates to clonal complexes (CCs) and SCC*mec* types was determined automatically by comparison of the hybridization profiles to previously typed reference strains [12].

2.5. Statistical analysis

Statistical analysis was done using SPSS 21 (IBM Inc. Chicago, IL). The χ^2 test was used to compare the significance of resistance profiles between MRSA and MSSA. Binary logistic regression was used to determine the association between MRSA nasal carriage, sex, and age. A P value less than 0.05 was considered statistically significant.

3. Results

During the study period, a total of 50 *S. aureus* isolates were isolated from the anterior nares of the 150 volunteers included in the study, resulting in a nasal carriage rate of 33.3%. Out of the 50 *S. aureus* isolates, 36 (72.0%) were obtained from men and 14 (28.0%) from women. The mean age of carriers was 22.2 years (SD 2.8; [19–38] y, median 22 y). Nineteen of the 50 *S. aureus* isolates (38.0%) were identified as MRSA: 14/36 (38.9%) in men and 5/14 (35.7%) in women. Therefore, the prevalence of nasal MRSA carriage among Kabul University students was 12.7% (19/150). MRSA frequency did not significantly differ by sex ($P = 0.84$) or age ($P = 0.18$).

Antimicrobial resistance rates of MRSA, MSSA, and *S. aureus* isolates are presented in Table 1. Most *S. aureus* isolates were resistant to penicillin (96%). The rates of resistance for other antibiotics were 32% for erythromycin, 8% for clindamycin, 18% for norfloxacin, 30% for kanamycin, 22% for tobramycin, 6% for gentamicin, 18% for cotrimoxazole, 4% for fusidic acid, 2% for chloramphenicol, and 8% for fosfomycin. The resistance rates of MRSA isolates to common

Table 2
Antimicrobial resistance genes identified in MRSA isolates from nasal carriers among non-medical students in Kabul, Afghanistan, between March and July 2018 (n = 19).

Antibiotic class	Resistance gene	All isolates n (%)
Beta-lactam	*bla*Z	17 (89.5)
	*mec*A	19 (100)
	*mec*C	0 (0)
Macrolide/lincosamide	*erm*A	0 (0)
	*erm*B	0 (0)
	*erm*C	2 (10.5)
	*lnu*A	0 (0)
	*msr*A	3 (15.8)
	*mph*C	3 (15.8)
Aminoglycoside	*aac*A-*aph*D	2 (10.5)
	*aad*D	1 (5.3)
	*aph*A3	3 (15.8)
	sat	3 (15.8)
Cotrimoxazole	*dfr*A	10 (52.6)
Fusidic acid	*fus*B	0 (0)
Tetracycline	*tet*K	9 (47.4)
	*tet*M	0 (0)
Chloramphenicol	*cat*	0 (0)
	cfr	0 (0)
	*fex*A	0 (0)
Mupirocin	*mup*A	0 (0)
Fosfomycin	*fos*B	3 (15.8)
Vancomycin	*van*A	0 (0)
Multidrug (efflux pump)	*sdr*M	4 (21.1)

antimicrobials were not significantly different from those of MSSA isolates (Table 1).

Moreover 36.8% (7/19) of MRSA isolates and 25.8% (8/31) of MSSA isolates were multidrug-resistant, i.e. resistant to at least 3 classes of antibiotics. All MRSA isolates were susceptible to linezolid, rifampicin, and fusidic acid.

Various resistance genes were detected in the 19 MRSA isolates: *bla*Z, *dfr*A, *and tet*K were the most frequently detected genes in 89.5%, 52.6%, and 47.4% of MRSA isolates, respectively. No strain harboured the resistance genes to fusidic acid (*fus*B*)*, chloramphenicol (*cat, fex*A), linezolid (*cfr*), mupirocin (*mup*A), or vancomycin (*van*A*)*. Of the 19 isolates phenotypically resistant to methicillin, all harboured the *mec*A gene, and no isolate was positive for *mec*C (Table 2). Concerning the staphylococcal chromosome cassette *mec* (SCC*mec*) element of these MRSA isolates, 18 (94.7%) harboured SCC*mec* type IV, and one (5.3%) harboured SCC*mec* type V.

Regarding the molecular typing of MRSA isolates, high clonal diversity was observed. The 19 MRSA isolates belonged to four different clonal complexes (CCs) and were assigned to seven distinct clones. One CC (CC22) dominated and accounted for 79% (15/19) of the isolates, and three other CCs (CC5, CC30 and CC88) accounted for 21% (Table 3).

The most frequently detected clone was CC22-MRSA-IV, carrying the toxic shock syndrome toxin (TSST-1) gene, which accounted for 63.2% (12/19) of MRSA isolates. Three resistance genes were detected in these isolates: the *bla*Z operon in 10/12 (83.3%), *dfr*A in 8/12 (66.7%), and *tet*K in 6/12 (50%). The enterotoxin gene cluster (*egc*; enterotoxins *seg, sei, sem, sen, seo,* and *seu*) was detected in all these isolates. The only other clone detected in at least 2 isolates was CC22-MRSA-IV positive for Panton-Valentine leukocidin genes, which accounted for 10.5% (2/19) of MRSA isolates. These two isolates were resistant to multiple antibiotics including aminoglycosides and erythromycin and harboured various resistance genes: the *bla*Z operon, *erm*C, *aac*A-*aph*D, and *dfr*A.

Only single isolates from 5 other MRSA clones completed this list, which represented 26.5% of all MRSA isolates and included well-known clones such as the Paediatric clone CC5-MRSA-IV TSST-

Table 3
Distribution and characteristics of the 19 MRSA isolates collected from anterior nares of non-medical students in Kabul University between March and July 2018 (n = 19).

Clonal complex (number of isolates)	Clone assignment (Alere)[a]	n	% of isolates	*agr* type	PVL	TSST	Enterotoxin genes	Antibiotic resistance genes	Antibiotic resistance
CC5 [1]	CC5-MRSA-IV TSST-1+, Paediatric clone	1	5.3	II	Neg	Pos	*sec, seg, sei, sel, sem, sen, seo, seu*	*bla*Z, *msr*A, *mph*C, *aph*A3, *sat*, *fos*B, *sdr*M	PEN, FOX, ERY, KAN
CC22 [15]	CC22-MRSA-IV, UK-EMRSA-15/Barnim EMRSA-like MRSA	1	5.3	I	Neg	Neg	*seg, sei,sem, sen, seo, seu*	*bla*Z, *tet*K	PEN, FOX, NOR
	CC22-MRSA-IV PVL+	2	10.5	I	Pos	Neg	*seg, sei,sem, sen, seo, seu*	*bla*Z, *erm*C, *aac*A-*aph*D, *aad*D (1/2), *dfrA*	PEN, FOX, ERY, KAN, TOB, GEN, NOR (1/2), FOF (1/2)
	CC22-MRSA-IV TSST-1+	12	63.2	I	Neg	Pos	*seg, sei,sem, sen, seo, seu*	*bla*Z (10/12), *dfr*A (8/12), *tet*K (6/12)	PEN, FOX, ERY (2/12), CLI (1/12), NOR (1/12), SXT (2/12)
CC30 [2]	CC30-MRSA-IV PVL+, Southwest Pacific Clone	1	5.3	III	Pos	Neg	*seg, sei,sem, sen, seo, seu*	*bla*Z, *msr*A, *mph*C, *aph*A3, *sat*, *fos*B, *sdr*M	PEN, FOX, ERY, CLI, KAN, NOR, CHL
	CC30-MRSA-V/VT PVL+, WA MRSA-124	1	5.3	III	Pos	Neg	*seg, sei,sem, sen, seo, seu*	*bla*Z, *msr*A, *mph*C, *aph*A3, *sat*, *tet*K, *fos*B, *sdr*M	PEN, FOX, ERY, KAN, NOR, SXT
CC88 [1]	CC88-MRSA-IV PVL+	1	5.3	III	Pos	Neg	*sep, sek, seq*	*bla*Z, *tet*K, *sdr*M	PEN, FOX
Total		19	100.0						

[a] MLST and clone assignments correspond to the comparison of the hybridization profiles on the DNA microarray to previously typed MLST reference strains [12].CHL, chloramphenicol; CLI, clindamycin; ERY, erythromycin; FOF, fosfomycin; FOX, cefoxitin; GEN, gentamicin; KAN, kanamycin; NOR, norfloxacin; PEN, penicillin; PVL, Panton-Valentine Leukocidin; SXT, trimethoprim-sulfamethoxazole; TOB, tobramycin; TSST-1, toxic shock syndrome toxin 1.

1-positive, the Southwest Pacific clone CC30-MRSA-IV PVL-positive, and the Barnim clone CC22-MRSA-IV UK-EMRSA-15 (Table 3).

Overall, 5/19 MRSA isolates (26.3%) were PVL-positive and 13/19 (68.4%) TSST-1-positive.

4. Discussion

This original study is the first molecular characterization of MRSA isolates in the Afghan community, describing the various clones circulating among young healthy individuals. A wide clonal diversity of MRSA was detected with four different clonal complexes, along with seven distinct clones. Molecular analyses showed a predominance of one MRSA clone, representing 63.2% of the isolates: the CC22-MRSA-IV TSST-1-positive. The six other clones detected were present only in one or two isolates.

The main limitation of this study is that only MRSA isolates collected from nasal carriers among Kabul university students were included. Therefore, the results do not reflect the entire epidemiology of MRSA nasal carriage in Afghanistan. Nevertheless, students are a population without risk factors, and the population in our study came from different parts of Afghanistan, therefore representing quite well the Afghan community. In addition, the diversity of strains highlighted by the typing results illustrates the position of Kabul as a centre of intense international exchange, as already described [7].

Results of the present study show that the epidemiology of MRSA carriage in the Kabul community is quite similar to the epidemiology of MRSA clinical isolates in Kabul hospitals. Of the 7 MRSA clones detected here, 5 clones accounting for 89.5% of the isolates were also reported in Kabul hospitals [7]. This suggests direct transfer between community and nosocomial MRSA clones in Kabul and diffusion of MRSA between these two settings.

The prevalence of nasal colonization with *S. aureus* in our study was 33.3%, a higher prevalence than that observed (19.6%) among healthy students of medical and non-medical universities in Urmia, Iran [13]; however, the prevalence of nasal carriage of MRSA was similar (around 13%). Another study in China, among medical students, showed a *S. aureus* nasal colonization rate of 15.4% and a prevalence of MRSA carriage of 3% [14], rates which are lower than our findings. In our study, the frequency of MRSA nasal carriage was not significantly different between male and female students, suggesting that sex may not be a risk factor for MRSA nasal colonization. However, a study from Iran [13] reported that male sex was a risk factor, and the authors argued it may be related to personal hygiene, especially by females. As previously reported, these data indicate that factors influencing MRSA colonization vary significantly, from country to country. Taken together, these results demonstrate that local epidemiological data are needed to guide national public health policies.

In the present study, three different MRSA clones were identified within CC22. This is a common and widespread clonal complex, and different MRSA lineages have emerged from this genetic background. The CC22-MRSA-IV TSST-1-positive was the most prevalent clone, representing more than half of all MRSA isolates. In a previous study about MRSA prevalence in clinical samples in Kabul health facilities, this clone was less frequent in hospital infections, observed in only 16.9% of MRSA isolates [7]. This strain is mostly reported in Saudi Arabia and the Middle East region [15]. Its presence in Afghanistan could be related to the large movement of Afghans to Saudi Arabia for pilgrimage each year.

The second most frequent MRSA clone was the CC22-MRSA-IV PVL-positive, which accounted for 7.7% of MRSA clinical isolates in our previous study in Kabul [7]. This clone has also been described in clinical samples from Western Europe (Germany, England, Ireland), Australia, Hong Kong, and the United Arab Emirates [16]. Only one isolate (5.3%) belonged to the CC22-MRSA-IV UK-EMRSA-15 Barnim clone; this is a common and pandemic strain of MRSA that has been found mainly in Western Europe but also in other parts of the world, including some Persian Gulf countries [17].

Two MRSA clones were identified within CC30: the CC30-MRSA-IV PVL-positive Southwest Pacific clone and the CC30-MRSA-V/VT PVL-positive clone. These lineages were also reported from Iran, United Arab Emirates, Saudi Arabia, Taiwan, Egypt, and Europe, both from the community and hospitals [15,16]. One MRSA clone was identified within CC5, the CC5-MRSA-IV TSST-1-positive Paediatric clone. This MDR clone, which also harboured the epidermal cell differentiation inhibitor gene (*edin*A), was first reported in hospitals in neonatal patients in Portugal [18] and in neonates and adults in Brazil [19] and Turkey [20]. Finally, one clone was identified within CC88, the CC88-MRSA-IV PVL-positive, which harboured enterotoxin genes and resistance genes only to penicillins and tetracyclines; this clone has been already reported in healthcare-associated infections in the region of Afghanistan [7], Saudi Arabia [15], and Kuwait [21].

In this study, PVL genes were found in 26.3% of MRSA isolates (n = 5), which is similar to another study among healthy adults in Iran [22]. This quite high prevalence of PVL genes among MRSA isolates should require more attention. The transmission of such virulent strains in universities and other crowded places where people are often in close physical contact can contribute to the transmission and spread of these strains among the community.

In conclusion, the relatively high frequency of MRSA nasal carriage, especially the virulent TSST-1 and PVL positive clones, among a healthy population in Kabul is a reason for concern, since individuals who harbour these isolates can act as reservoirs for outbreaks of MRSA infections. Supporting policies are necessary for preventing the hospital–community spread of MRSA and reducing nasal carriage through continued efforts to enhance hygiene among students.

Funding: This work was financially supported by Santé Publique France and the French Ministry of Foreign Affairs.

Competing interests: None declared.

Ethical approval: This study was approved by the Academic Council of the Faculty of Pharmacy and Research Committee of Kabul University with approval numbers FoP.124, 12/02/2018 and KURC.115, 20/02/2018. Written consent from all participants was obtained.

Acknowledgements: The authors would like to thank Christine Gardon and all the technicians of the French National Reference Centre for staphylococci for technical assistance.

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